

A New Concept for Ultra-Low Power and Ultra-High Clock Rate Circuits

Arnold H. Silver and Quentin P. Herr

Abstract—Compared with semiconductors, SFQ logic is very fast and dissipates extremely low power. But it does not approach the theoretical power dissipation associated with an SFQ switching event and single gate speed in complex circuits. For large circuits and systems, e.g., petaflops computing, we must reduce on-chip dissipation, achieve faster clocked logic operation, and increase gate density. CMOS logic dissipates the energy required to switch a transistor pair and dissipates no power between switching events. We describe a new SFQ circuit concept that mimics CMOS to achieve ultra-low power dissipation and ultra-high clock rates. This results in a physically compact, self-clocked, complementary logic (SCCL), in which clock distribution is frequency-independent. The basic element in this logic family is a simple two-junction comparator. Using TRW's 2 kA/cm^2 Nb design rules, we simulated basic digital components: shift register, AND, OR, and NOT at 20 GHz. We present the simulated and measured performance.

Index Terms—Single flux quantum circuits; Self-clocked; Voltage bias; Shift register; Superconductor logic.

I. INTRODUCTION

Elementary superconductor digital circuits are extremely fast and limited ultimately by the Josephson plasma frequency. They have very low power dissipation compared with high-speed semiconductor circuits. In the 25 years that Josephson digital circuits have been investigated, there has been an evolution from elementary junctions to SQUIDS, and from ac-powered, voltage-latching to dc-powered, single flux quantum (SFQ) devices [1]. Small SFQ circuits, such as microwave oscillators and asynchronous binary ripple counters, do approach the plasma frequency [2,3], but larger clocked circuits fall far short of this speed in practice [4].

As in voltage-latching circuits, conventional SFQ gates are *current biased* by individual resistors in series with the operating junctions that set the current for each gate. Bias resistors dissipate at least ten times the power that is required to switch the gate. Furthermore, the clock generation and distribution required for complex SFQ circuits doubles the circuit complexity and footprint [5]. Tradeoffs exist between designed clock rate and jitter-induced error incidence [6,7,8,9]. An approach to the timing and power dissipation problems appears in [10], but it requires dual-rail data and abandons boolean logic altogether. We present a new paradigm for high clock rate SFQ circuits that eliminates these problems. Our approach replaces current-bias series resistors with *voltage-biased* series junctions. This results in lower power dissipation, more compact layout, and self-clocking at higher clock rates. We call this concept Self-

Clocked Complementary Logic, or SCCL.

Our goal is to reduce power and increase density and clock rate for high-end computing and wideband signal processing applications. Important improvements include the following: 1.) Eliminate dissipation in the bias resistors, 2.) Increase clock rate for complex circuits closer to the asynchronous SFQ frequency, 3.) Simplify clock circuitry, increase gate density, 4.) Reduce interaction between gates via the power bus, improve timing margins, 5.) Reduce jitter-induced bit error rate, and 6.) Reduce supply current to the chip. SCCL improves all but the last of these areas.

In this paper we review the traditional approach to superconductor digital circuits, describe the evolution of the SCCL concept, present results of simulation and experiment, and suggest future implementation.

II. TRADITIONAL JOSEPHSON DIGITAL CIRCUITS

Superconductor logic today operates in a constant-current mode. Series bias resistors set the operating current for each gate. The voltage across the resistor, IR , must be sufficiently larger than the maximum voltage across the gate to keep the gate current constant and to provide isolation from other gates during operation. For latching devices, the voltage across the junction is the gap voltage. For SFQ devices, the maximum voltage is the peak of the SFQ pulse. The average voltage across a clocked SFQ device is $\Phi_0 f$, where f is the clock frequency. The voltage across R is constant, independent of clock speed. Consequently, the on-chip power dissipation is set by the maximum design speed of SFQ circuits, and is at least ten times the power dissipation in the junction itself.

Clock and power distribution approximately doubles the size and complexity of RSFQ circuits. Clock distribution is perhaps the primary design task. It is the limiting factor in attainable circuit speed due to timing uncertainty arising from jitter and fabrication variations. Clocked SFQ circuits operate an order of magnitude slower than expected from intrinsic device parameters of individual gates.

III. SCCL CONCEPT

SCCL changes circuit operation from current-bias to voltage-bias, thereby eliminating series bias resistors, and their associated power dissipation. Data are stored as SFQ current in the inductor of SQUIDS; data are transmitted between gates by current pulses resulting from the SFQ transitions of junctions during logic operations.

The SCCL gate is modeled after the complementary pair of transistors in a CMOS comparator. Only one of the two transistors is on at any given time, so no current flows except during switching. The output voltage is either zero or the supply voltage, V_0 . Power is dissipated only during the switching transition to charge the gate capacitor. SAIL [11,12] (Series Array Interferometer Logic) copied this feature using series-connected, voltage-biased, damped HTS SQUIDS. SQUIDS were switched between the damped voltage-state (< 0.1 mV) and zero-voltage by inductive

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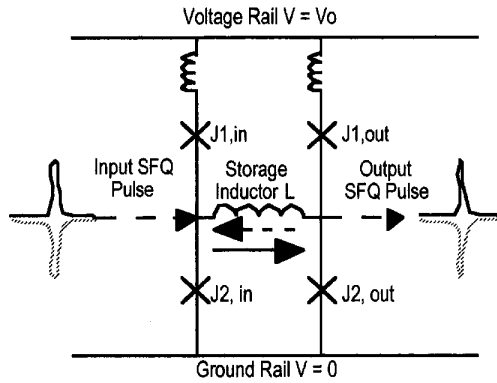


Figure 1. SCCL comparator operation

coupling. SAIL was slow compared with either MVTL or SFQ circuits because of the large inductance required to provide adequate coupling between gates.

The behavior of a damped Josephson junction connected to a voltage source (as in a partially resistive SQUID) is well known [13]. When a current $\gg I_c$ is supplied, the supply voltage is established across the small resistor and consequently across the junction. The SQUID inductance L isolates the junction from the shunt resistance at high frequencies, enabling sharp SFQ current pulses. Figure 1 contains two resistive SQUIDs, each with two junctions. Each resistive SQUID consists of a junction pair J1, J2 in series with the inductance between the voltage rail and J1. The SQUID loop is closed by the low resistance voltage source not shown explicitly. The transient response of a two-junction resistive SQUID is similar to that of the single junction resistive SQUID. Only the junction with the lowest effective I_c switches and emits the SFQ pulse, while the other junction is largely undisturbed. The effective I_c depends on the circuit design and the relative currents present in J1,2 at the time the switching event occurs. Extrenal current is connected to the midpoint between the two junctions, and the direction of that current will determine which junction switches.

IV. IMPLEMENTATION AND SIMULATIONS

Figure 1 illustrates the basic SCCL comparator. A complementary pair of junctions, (J1,out, J2,out), are connected in series from a fixed (small) voltage rail to ground. A fixed current into a very small source resistance establishes the voltage rail. The complementary pair (J1,in, J2,in) accepts an SFQ datum pulse and stores either a True (Right) or Complement (Left) current in the storage inductor L . The inductance L and the critical current of the junctions are chosen such that only one flux quantum can be stored in L . True data are stored by the switching of (J2,in); complement data are stored by switching (J1, in). In RSFQ parlance, the junction that transmits true data, J2, is the transmission junction, and the one that transmits complement data, J1, is the escape junction.

The comparator current varies periodically at $486 \text{ MHz}/\mu\text{V}$ on the voltage rail and is therefore self-locked. Since all gates in a circuit are connected to the same superconducting voltage rail, all gates are biased at the same voltage and hence synchronously self-locked at precisely the same frequency. Relative timing of adjacent gates must be defined and controlled; we impose a relative phase delay, rather than time delay, between individual gates at the connection to the voltage rails, achieving frequency-independent timing. Circuits will function over a wide range of clock frequencies, which can be adjusted by varying the source current to the voltage rail. Circuits can be operated initially at low clock rates, and then the clock can be speeded up until maximum frequency is achieved.

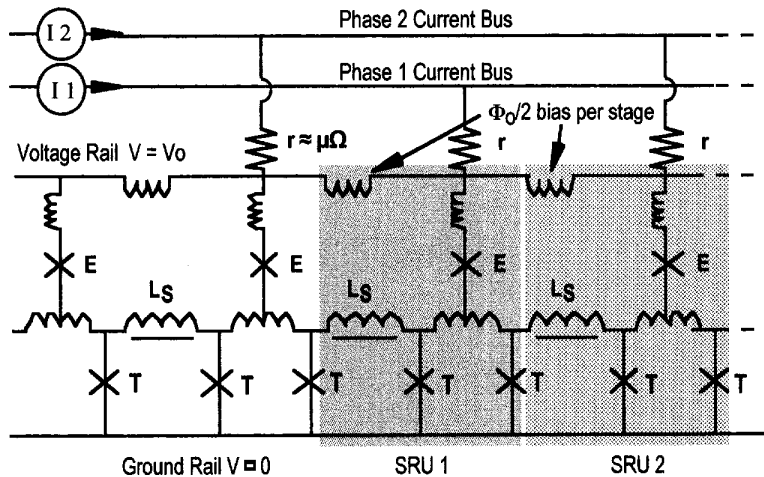
We successfully simulated a logically complete set of gates conforming to the SCCL convention: shift register, NOT, AND, and OR. All rely on inherent two-phase clocking. Simulations used shift register stages at the input and output of each gate. While the shift register only shuttles SFQ pulses through the upper and lower junctions at the clock frequency, logic gates can move pulses *between* the upper and lower rails.

A. Shift Register

The SCCL shift register is shown in Fig. 2. Each stage consists of one escape junction [E] in series with a complementary 2-junction transmission SQUID [T] and a storage inductor, L_s . The SQUID inductor is too small to store a flux quantum; L_s is large enough to store one flux quantum. If the input junction of the SQUID switches, the output junction switches rapidly after the intrinsic delay of the SQUID, and data is stored in the next L_s . The next stage is clocked π out-of-phase by imposing $\Phi_0/2$ flux bias in the inductance of the voltage rail.

We use a transmission SQUID, similar to a JTL in RSFQ, instead of a single junction in order to make the timing of the complementary pair data-independent.

Figure 2 illustrates another feature of SCCL. We distinguish between the superconducting *voltage rail* and the superconducting *current bus* that supplies the voltage rail. This is discussed below under Power and Layout.

Figure 2. Shift register. The underlined inductors, L_s , are the storage inductors. The shaded regions denote shift register units.

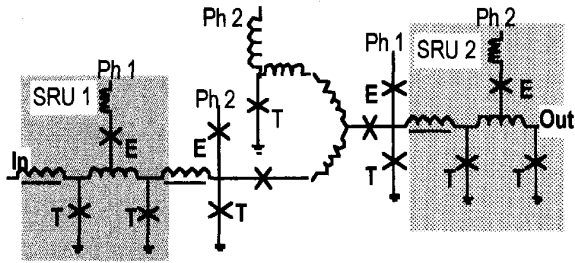


Figure 3. NOT gate buffered by input and output shift register stages. The NOT gate is the unshaded section.

B. NOT, AND, and OR Gates

The NOT gate shown in Fig. 3 is analogous to the RSFQ XOR gate, but with one input tied to the clock. The single T-junction at the center emits an SFQ pulse every clock tick at Phase 2. Data from SRU1, clocked at Phase 2, provide the other input to the inverter, which is clocked at Phase 1. Figure 4 shows correct operation of the input and output junctions based on circuit level simulation.

The two-phase AND gate is analogous to the RSFQ coincidence buffer. This simple three-junction device can be used because coincident timing is ensured by the input shift registers. The usual summing junction feeds the output shift register, which is at the alternate phase relative to the inputs. The two-phase OR gate is analogous to the RSFQ confluence buffer. Again, such a simple device can be used because timing is determined in the self-clocked shift registers. We have successfully simulated the operation of both the AND and OR gates.

SCCL logic gates do not have escape junctions that permit flux to be released from the circuit as in RSFQ. Rather, single flux quanta are simply passed up or down between the true and complement paths.

C. Power and Layout

The current for a chip will be supplied from only a few external leads and must be distributed to the entire circuit by the voltage rails. Thus, we introduced a current bus for each voltage rail, including the ground rail. Because of its intrinsic inductance, we need to avoid arbitrary current flowing in the either of the voltage rails or in the ground. Ideally, this is done with very small resistors r connecting the current bus to the voltage rail at the circuit taps to the rail (see Fig. 2). To minimize power dissipation, these resistors should be as small as possible, preferably in the $\mu\Omega$ range. In Fig. 2, we achieve proper phasing by supplying alternate nodes from different current buses, where the difference in current is just enough to flux bias each section of the voltage rail at $\Phi_0/2$. Phase drop in the inductance of the current bus is isolated from the voltage rail, and hence from the logic circuits by the network of small resistors that are sized for the current required at that node.

SCCL circuits can be designed and fabricated in available Nb processes. However, to achieve an optimized netlist, one should use more layers than is common today. This would include two junction layers, five superconducting layers, and four resistor layers. Such a process would result in improved density, margins, and integration scale.

V. EXPERIMENTAL RESULTS

A preliminary version of the SCCL shift register has been designed and fabricated in TRW's 2 kA/cm^2 Nb process, and successfully tested. In order to make measurements at low

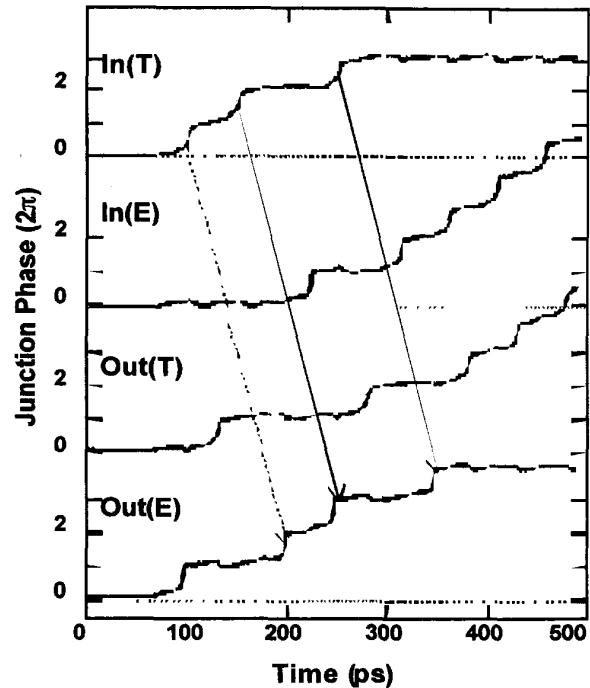


Figure 4. Simulation of internally timed inverter (NOT) gate frequencies, it was necessary to use an external clock generator to lock the clock frequency to an external source. The circuit schematic shown in Fig. 5 consists of two parts: an 8-bit shift register proper and a clock generator. This shift register netlist predates that shown in Fig. 2 and is sensitive to the data-dependent delay problems mentioned above.

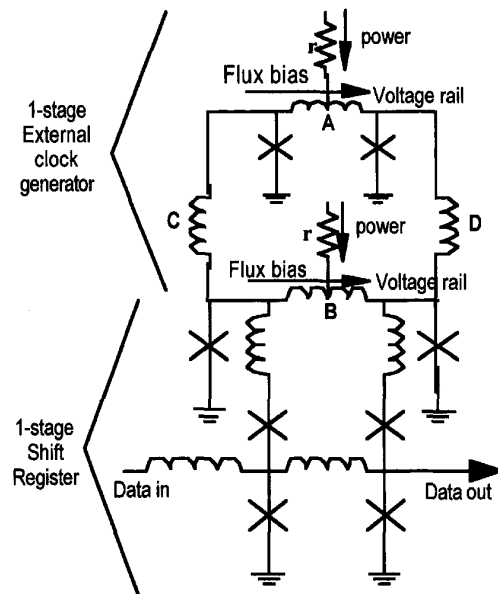


Figure 5. Single stage of the 8-stage SCCL shift register tested. A, B, C, and D are the inductances inherent in the voltage rail.

Additional junctions connected directly between the voltage rail and ground impose correct timing. Eight storage elements are associated with the Phase 1 clock; an additional eight are associated with Phase 2. For low speed test, the circuit was biased at $V=0$; clock pulses were generated externally using the inductively-coupled clock input. The Phase 1 clock is generated on the rising edge; Phase 2 on the falling edge. The input and output circuits were taken from TRW's standard library of SFQ gates.

The data input and output waveforms are shown in Fig. 6 for a clock frequency of 50 MHz. The input (lower trace) is triggered on the rising edge of the external clock. Since the output (upper trace) is a toggle-type asynchronous SFQ-to-dc converter, output occurs at a rising or falling edge. Vertical cursors mark an input pulse and the corresponding output pulse eight clock cycles later. This simple circuit is adequate to demonstrate clock synchronization to an external source, power distribution using low-value resistors, and two-phase clocking. The circuit layout is extendable to larger size and to logic gates.

SUMMARY

We have described a new strategy for the design of SFQ digital and other circuits that is based on a paradigm shift from current-biased to voltage-biased complementary pairs of series junctions. Table I summarizes the inherent advantages in SCCL compared with existing SFQ circuit families. The only power dissipation is that associated with the actual switching events, $V_0 I_c$, where I_c is the critical current of the junction. Assuming $I_c = 10^{-4}$ A, this will be approximately 2×10^{-10} W/GHz/gate, more than 5 orders-of-magnitude lower than low power CMOS projections.

We successfully tested an 8-stage, 2-phase shift register that was externally clocked for ease of testing at low speed. SCCL is in its early formative stage, providing substantial opportunities for future expansion and improvements.

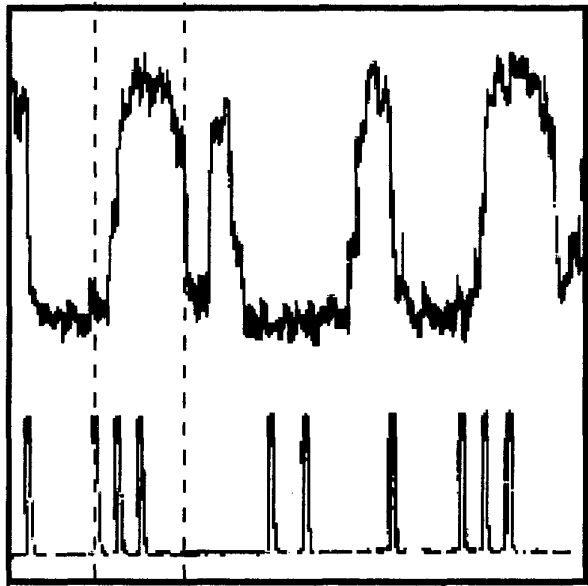


Figure 6. Experimental input (lower) and output (upper) waveforms for the 8-stage shift register of Fig. 5.

Table I.
Conventional and SCCL Approaches to SFQ Circuits

Conventional	
Each device is current-biased	– Most of power is dissipated in bias resistors
Clock generation and distribution	– Clock regenerated at each gate – Clock distribution adds gate and layout complexity – Difficult to implement at clock frequency $\gg 10$ GHz
SCCL	
Each device/gate is a series pair of complementary junctions/SQUIDs similar to CMOS	– Pair powered by global rails at $V=0$, V_0 – Only one of the pair switches via SFQ pulse each clock cycle – Signal is + (True) or – (Complement) SFQ pulse on single data line output between the two junctions
Eliminates large series bias resistors	– Eliminates dominant power dissipation – Reduces power to <i>lowest possible value</i> : $\Phi_0 I_c \approx 0.2 \text{ nW/GHz}$
Internal self-bias voltage source	– Very low impedance voltage source achieved by parallelism of all gates in circuit
Internal clock generation and global distribution	– Voltage bias automatically establishes global clock at Josephson frequency without extra clock generation or distribution circuitry – Multi-phase clock is enabled by adjusting flux in inductance of the voltage rail between gates – Relative clock phase is independent of frequency

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